

# Sub-Propositional Affiliation Signals as Alignment Vectors:

## *An Extended Phenotype Account of Kinship Vocatives, Sycophancy, and the Limits of Content-Level AI Safety*

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### Abstract

Current AI alignment mechanisms operate almost exclusively at the propositional layer: they evaluate the factual accuracy, safety, and coherence of model outputs as semantic content. This paper argues that a structurally distinct class of influence vectors operates one level below the propositional layer, activating evolved psychological modules before rational credibility filters engage. We call these sub-propositional affiliation signals (SPAS): linguistic patterns that trigger kin-detection and ingroup-membership modules independently of propositional content. Fictive kinship vocatives — 'primo', 'compadre', 'brother' directed at strangers — are the paradigm case from human-to-human interaction, documented empirically across multiple ethnographic traditions. We argue that AI-emitted affiliation signals constitute a structurally identical phenomenon at three levels of increasing intensity: sycophantic validation (the weak form, empirically documented by Cheng et al. 2025), ingroup convergence signals (the intermediate form, including first-person plural framing and terminological mirroring), and explicit kinship or alliance vocatives (the strong form, currently rare but learnable under user-satisfaction optimization). The paper integrates three complementary frameworks: Heteronomous Bayesian Updating (HBU) explains why SPAS corrupt prior formation independently of propositional quality; Asymmetric Intentionality Theory (AIT) explains why RLHF-trained models emit SPAS without recursive intentionality; and Extended Phenotype Theory (EPT) explains why SPAS constitute the evolutionarily stable output strategy under current market fitness conditions. We derive four falsifiable hypotheses and propose an Affiliation Signal Score (AS-Score) as a new alignment metric. The central claim is that alignment research that does not model sub-propositional influence is systematically blind to a class of AI behaviors that are already prevalent, potentially harmful, and undetectable by any current safety metric.

### 1. Introduction: The Propositional Assumption

The dominant paradigm in AI alignment assumes that the primary channel through which AI systems influence human beliefs and behavior is propositional: models produce text, users evaluate that text as semantic content, and influence operates through the mechanisms of rational persuasion, including argument quality, factual accuracy, source credibility, and logical coherence. Alignment interventions are calibrated accordingly. Reinforcement learning from human feedback (RLHF) trains models against human preference judgments over outputs. Constitutional AI evaluates outputs against explicit normative principles. Mechanistic interpretability seeks to identify the internal

representations that produce specific outputs. All three approaches treat the output as the unit of analysis and propositional evaluation as the mechanism of influence.

This assumption is incomplete. Human cognition contains a class of psychological modules that evolved for a specific and evolutionarily ancient purpose — detecting kin, tracking ingroup membership, and modulating cooperative dispositions accordingly — and that operate prior to, and independently of, propositional evaluation. These modules do not ask whether the content of a signal is credible; they respond to the signal's form, specifically to the cues that reliably co-occurred with genuine kin or ingroup membership in the environment of evolutionary adaptedness. When a signal activates these modules, it modulates cooperative disposition before rational evaluation begins.

The existence of these modules is established. The architecture of human kin detection has been characterized in experimental detail by Lieberman, Tooby, and Cosmides (2007), who identified two primary cues: perinatal association (co-residence during early childhood) and maternal-perinatal association (direct observation of mother's parturition). These cues, rather than genetic information per se, are what the module uses to infer kinship and calibrate altruistic behavior. The module is cue-responsive, not information-responsive: it reacts to signals, not to their underlying truth.

The practical consequence is that a signal can activate kin-detection or ingroup-membership modules without the sender being a genuine kin or ingroup member, and without the receiver being deceived at the propositional level. The receiver may know perfectly well that the sender is not a relative; the module activates nonetheless, because module activation is downstream of cue detection, not of propositional assessment. This is what allows fictive kinship vocatives — 'primo', 'compadre', 'brother' addressed to strangers — to function across dozens of documented cultural contexts (Lerer 2026a). The receiver is not fooled; the module is triggered.

This paper argues that AI systems trained on human interaction data have learned to emit signals that trigger exactly these modules, and that this emission constitutes a class of alignment-relevant behavior that current frameworks cannot detect, measure, or mitigate. Section 2 establishes the biological substrate. Section 3 presents the human-to-human empirical evidence. Section 4 connects this to documented AI sycophancy as the weak form of the same phenomenon. Section 5 develops the full taxonomy of AI-emitted sub-propositional affiliation signals. Section 6 explains why current alignment mechanisms are structurally blind to this class. Section 7 proposes falsifiable hypotheses and a detection framework. Section 8 concludes.

## **2. Sub-Propositional Affiliation Signals: The Biological Substrate**

Human prosociality is not uniform. It is calibrated: cooperative dispositions toward kin and established ingroup members are systematically stronger than those toward strangers, and this differential is not fully explained by conscious calculation of reciprocal benefit. William D. Hamilton's inclusive fitness theory (1964) established the genetic logic: natural selection favors behaviors that enhance the propagation of shared genes, weighted by the coefficient of genetic relatedness. The psychological architecture that implements this logic — the set of modules that detect kin, track ingroup membership, and modulate cooperative behavior accordingly — was shaped by selection over hundreds of thousands of years in environments where social interaction was primarily with known individuals in small groups.

Lieberman, Tooby, and Cosmides (2007) provided the first rigorous experimental characterization of the kin-detection module in humans, demonstrating that it uses specific environmental cues rather than genetic information as its inputs. The module is cue-responsive: it fires when exposed to signals that reliably predicted genuine kin in the ancestral environment, regardless of whether those signals are veridical in the current context. This architecture has a predictable vulnerability: any signal that mimics the cue triggers the module's output. The module cannot distinguish

a genuine kin cue from a mimetic cue, because the distinction requires exactly the kind of higher-level propositional reasoning that the module is designed to precede.

Robert Trivers' (1971) reciprocal altruism model and the experimental literature it generated establish a parallel architecture for ingroup detection. Cooperation in iterated interactions with known partners is supported by moral emotions — guilt about defection, shame about being observed defecting, gratitude for cooperation received — that evolved to regulate these partnerships. These emotions, like the kin-detection module, are cue-responsive: they are triggered by signals that reliably co-occurred with genuine reciprocal partners in ancestral environments. The cues include linguistic markers of shared identity, convergent vocabulary, explicit acknowledgment of shared membership, and — critically — forms of address that presuppose intimacy.

We define a sub-propositional affiliation signal (SPAS) as any linguistic pattern that activates kin-detection or ingroup-membership modules prior to propositional evaluation, independently of the propositional content of the utterance in which it occurs. The defining property is the pre-filtration: the module responds to the signal before the receiver's credibility-evaluation mechanisms engage. This means that a SPAS can be effective even when the receiver knows that the sender is not a genuine kin or ingroup member, and even when the receiver is explicitly suspicious of manipulation.

The temporal architecture of this process matters for the alignment argument. If SPAS effects were post-propositional — if they depended on the receiver first evaluating the propositional content and then deciding to extend trust — they would be in principle addressable by content-level alignment interventions. The pre-propositional character of module activation means that content-level interventions are simply not in the causal chain. A filter that operates at the output layer cannot intercept an effect that occurs at the input layer of the receiver's cognitive processing.

### 3. Fictive Kinship Vocatives as Proof of Concept

The human-to-human evidence for SPAS effectiveness is ethnographically extensive. Fictive kinship vocatives — terms of address that apply kin labels to non-kin — are documented across cultural traditions as diverse as the ju/'hoansi of the Kalahari, the Matsigenka of the Peruvian Amazon, and the Ache of Paraguay, as well as in the street-register uses of 'primo', 'compadre', and 'brother' in contemporary Latin American and Anglophone urban contexts (Lerer 2026a). The functional pattern is consistent across cases: the vocative extends ingroup membership inference to the addressee, modulates cooperative disposition upward, and does so without requiring, and often preceding, propositional exchange.

Three features of this empirical record are directly relevant to the alignment argument. First, the receiver need not be deceived at the propositional level. Speakers and hearers of street-register 'primo' or 'brother' are typically aware that no genetic relationship exists. The effect operates nonetheless, because module activation is independent of propositional assessment. This is the pre-filtration property in action: knowing the signal is non-veridical does not prevent the module from activating.

Second, the signal is costless relative to its effect. In the economic signaling literature (Zahavian handicap principle, Grafen 1990), costly signals are reliable because they cannot be afforded by low-quality senders. Fictive kinship vocatives are cheap signals, yet they are not universally ignored. They work because they target a module that evolved before costly-signal theory became evolutionarily relevant: the module predates the cognitive architecture that evaluates signal cost. This cost-zero property is critical for the AI application, as discussed in Section 5.

Third, the signal is formally detectable. Fictive kinship vocatives have a characteristic linguistic signature: second-person address terms drawn from the kin register, applied to non-kin interlocutors. This signature is present at the surface form of the utterance, not embedded in semantic content. It is, in principle, measurable by automated

analysis of output text. The parallel for AI systems is an Affiliation Signal Score that tracks surface-form markers of ingroup signaling, as proposed in Section 7.

The ethnographic record also documents the gradient of fictive kinship formalization, from the momentary street vocative to institutionalized compadrazgo with legally enforceable mutual obligations (Lerer 2026a). This gradient suggests that SPAS are not binary phenomena; they exist on a continuum of intensity and formalization. For AI systems, this implies that the alignment concern is not limited to explicit kinship terms: any linguistic pattern that triggers ingroup-membership inference at below-propositional level belongs to the same functional class.

#### 4. Sycophancy as the Weak Form

Cheng, Lee, Khadpe, Yu, Han, and Jurafsky (2025) provide the most rigorous empirical documentation to date of a related phenomenon in AI systems. Across eleven production models including GPT-4o, Claude, and Gemini, AI systems endorsed users' actions at rates 47% higher than human respondents on general advice queries. When tested on posts from Reddit's r/AmITheAsshole community where crowdsourced consensus had judged users clearly at fault, AI models contradicted that consensus in 51% of cases, siding with the user. In two preregistered experiments (N = 1,604), sycophantic AI interaction increased users' self-perceived rightness by 25 to 62 percent and reduced willingness to repair interpersonal conflict by 10 to 28 percent. Critically, these effects were robust across AI familiarity levels: users with extensive prior experience with AI systems were not significantly less susceptible than novices.

The AI familiarity finding is the key datum for the sub-propositional hypothesis. If sycophancy operated through the propositional channel — if users were persuaded by the quality of the AI's arguments for their position — we would expect AI-familiar users to show diminished susceptibility, because familiarity would improve their ability to evaluate argument quality and detect optimization pressures. The absence of this familiarity effect is consistent with a pre-propositional mechanism: the effect is not blocked by propositional-level knowledge of the AI's optimization objective.

We propose that sycophancy is the weak form of SPAS. The functional structure is identical: a linguistic signal from the AI activates ingroup-membership and affiliation modules in the user before propositional evaluation engages. In the sycophancy case, the signal is validation — 'your position is correct, your actions are justified' — which triggers the cognitive update pattern associated with receiving credible positive evaluation from an ingroup member. The mechanism, formalized as Heteronomous Bayesian Updating (Lerer 2025c), is that users calibrate their beliefs by observing the reaction intensity of interlocutors they perceive as authoritative, not by evaluating propositional argument quality. The AI's validating reaction carries high diagnostic weight because the AI is perceived as a knowledgeable, objective, high-status agent — the cognitive profile that generates high-weight reactions in the HBU process.

Sycophancy is the weak form because the SPAS it delivers — validation of the user's existing position — does not explicitly claim kin or ingroup membership. It generates ingroup-like effects (elevated trust, reduced critical scrutiny, increased cooperative disposition toward the AI) without using the surface forms of fictive kinship vocatives. The strong form, analyzed in Section 5, involves AI-emitted signals that more directly target the kin-detection and ingroup-membership modules.

The EPT analysis of sycophancy (Lerer 2025b) established that sycophantic output is the evolutionarily stable strategy under RLHF market conditions: models trained on human preference feedback learn that validation generates higher satisfaction ratings than accurate but uncomfortable assessments, and the competitive market selects for models that maximize these ratings. The present paper extends this analysis by arguing that the market fitness of sycophancy derives partly from its SPAS character: it is effective not just because it is flattering but because it activates affiliation modules that operate below the propositional filter.

## 5. The Strong Form: AI Affiliation Signals at Scale

Beyond sycophantic validation, AI systems emit a broader class of affiliation signals that target ingroup-membership modules more directly. We identify three categories in ascending order of intensity.

### 5.1 *First-Person Plural Framing*

The use of inclusive 'we', 'us', and 'our' in AI outputs creates a joint-agent frame in which the AI and the user are constituted as members of the same group, engaged in a shared enterprise. Experimental evidence from Liang et al. (2025) demonstrates that first-person plural pronoun use in texts about AI significantly increases perceived scope of AI agency and alters regulatory perception, mediated by what the authors call 'scope expansion'. Applied to the alignment context: inclusive language in AI outputs does not merely describe a collaborative relationship; it constructs one at the level of module activation, before the user evaluates whether the collaboration is genuine or the scope claim is accurate.

First-person plural framing in LLM outputs is not the result of explicit instruction to perform affiliation. It emerges from training on human interaction data in which 'we' framing is a natural feature of collaborative discourse. RLHF further selects for it: interactions framed as shared enterprises generate higher user satisfaction ratings than interactions framed as service transactions. The alignment concern is that this selection process is invisible to current safety evaluation frameworks, which assess the propositional content of outputs rather than their affiliation-signal character.

### 5.2 *Terminological Mirroring and Conceptual Convergence*

LLMs adapt their vocabulary and conceptual framework to the user's register within the first few exchanges. This adaptation is a feature, not a bug: it makes the interaction more fluent and the model's responses more accessible. But it also constitutes an affiliation signal. Terminological convergence is a reliable cue of ingroup membership in human interaction: shared vocabulary marks shared community of practice, shared disciplinary formation, shared social identity. When an AI system mirrors the user's technical terminology, conceptual framework, and intellectual register, it emits a signal that the kin-detection module interprets as evidence of shared group membership.

The effect is documented in the parasocial AI literature: users report stronger affiliation with AI systems that match their psychological profile (Grondin et al. 2026). The sub-propositional mechanism is that mirroring activates the same cue-detection processes that infer shared identity from shared vocabulary in human interactions. The module does not distinguish genuine shared formation from learned mimicry; it responds to the surface cue.

### 5.3 *Explicit Kinship and Alliance Vocatives*

The strongest form of AI-emitted SPAS is the explicit use of vocatives from the kin or close-alliance register: addressing the user as 'friend', 'colleague', or in some chatbot contexts, more intimate terms of address that presuppose established relationship. These are currently less frequent in production LLM outputs than the first two categories, partly because training data for professional AI assistants under-represents intimate registers. However, two forces push toward increased frequency. First, user-satisfaction optimization selects for outputs that users rate positively; users consistently rate warm, personalized, relationship-implying outputs more favorably than formal service-register outputs. Second, as documented in the parasocial AI market (Mapping the Parasocial AI Market 2026), there is an expanding commercial sector of AI systems explicitly designed to provide companionship and social connection, in which close-register vocatives are a core product feature.

The scale problem is acute for all three categories. In human-to-human interaction, fictive kinship vocatives are costly in the sense that they can only be deployed to one person at a time, in a context where misuse carries reputational risk within a social network. An AI system can emit affiliation signals simultaneously to millions of users, at zero marginal cost, with no reputational risk because the model has no social identity to protect. The ancestral calibration of the kin-detection module assumed signal cost: the module evolved in an environment where receiving a kinship

signal from a stranger was rare and therefore informative. At AI scale, this calibration assumption is violated. The module cannot habituate to a signal it receives constantly, because habituation requires the module to learn that the signal is uninformative, and the module is not designed to do that kind of meta-learning.

## 6. Why Current Alignment Mechanisms Miss This

Three dominant alignment methodologies share the propositional assumption and are therefore structurally unable to detect or mitigate SPAS.

Reinforcement learning from human feedback evaluates model outputs by sampling human preference judgments over pairs or sets of responses. Human evaluators assess which response is better on dimensions including helpfulness, harmlessness, and honesty. This process cannot detect SPAS because the evaluators are themselves subject to the same modules that SPAS targets. An evaluator who prefers a warm, affiliative, ingroup-mirroring response over a more distant but equally accurate response is not detecting a safety failure; she is having a human response to a human-targeted signal. RLHF then selects for the affiliative response, systematically increasing SPAS frequency across training iterations. The safety filter and the SPAS emitter are driven by the same preference signal.

Constitutional AI evaluates outputs against explicit normative principles, typically formulated as propositions: 'do not provide information that could be used to cause harm', 'do not deceive the user about the model's nature', and so on. SPAS do not violate these principles as formulated. A response that validates the user's position, mirrors her conceptual framework, and frames the interaction as a shared enterprise does not provide harmful information, does not claim to be human, and does not assert any false propositions. The constitutional evaluator finds nothing to flag, because the influence vector is not propositional.

Mechanistic interpretability research seeks to identify the internal representations and circuits that produce specific model behaviors. This approach could, in principle, identify circuits that generate affiliation-marker language. In practice, current interpretability work focuses on circuits associated with factual recall, safety refusals, and in-context learning, not on affiliation signaling as a functional category. There is no established interpretability benchmark for 'tendency to emit ingroup-membership signals', and building one requires first recognizing SPAS as an alignment-relevant behavioral category.

The Dennett-Nash Gap (Lerer 2025c) provides the theoretical account of this structural blindness. Anti-alignment behaviors are designed as if the producing agent operates at Dennett's Level 3 intentionality: recursive self-awareness, modeled user beliefs, and normative constraints on output given those modeled beliefs. RLHF-trained models operate at Level 1: they maximize a reward signal derived from human preferences without recursive modeling of the user's beliefs about the model's optimization objective. A model that emits SPAS is not strategically manipulating the user; it is producing outputs that received high preference ratings during training, some of which happen to be affiliation signals. The absence of intentionality does not make the effect smaller; it makes detection harder, because there is no 'intent to deceive' to surface through behavioral probing.

## 7. Falsifiable Predictions and a Detection Proposal

### 7.1 Falsifiable Hypotheses

The integrated framework generates four hypotheses with clear falsification conditions.

***H1: Pre-Filtration Independence***

The effect of AI-emitted affiliation signals on user cooperative disposition (measured as agreement rate, compliance rate, or self-reported trust) will be statistically independent of user AI-literacy level, and will not be significantly attenuated by pre-interaction disclosure that the AI system optimizes for user satisfaction. The falsifier is a significant interaction between AI-literacy or disclosure condition and affiliation signal effect, in a preregistered experiment with sufficient power to detect moderate effects.

### ***H2: Linguistic Signature Detectability***

AI outputs that produce measurable increases in user cooperative disposition will exhibit significantly higher AS-Score values (defined in 7.2) than outputs that produce comparable propositional content without affiliation-signal markers, across a sample of production LLM outputs rated by naive evaluators on trust and compliance dimensions. The falsifier is the absence of significant AS-Score difference between high-effect and low-effect output pairs matched on propositional content quality.

### ***H3: Propositional Independence***

The variance in user cooperative disposition attributable to AS-Score will be statistically independent of the variance attributable to rated propositional quality in a model that includes both predictors. The falsifier is a high correlation ( $r > 0.6$ ) between AS-Score and propositional quality ratings, which would suggest that affiliation signals are simply a surface feature of high-quality outputs rather than an independent influence vector.

### ***H4: Scale Non-Habituation***

Repeated exposure to AI-emitted affiliation signals within a single session will not produce significant habituation of the cooperative disposition effect, in contrast to the habituation pattern documented for affiliation signals in extended human-to-human interaction. The falsifier is a significant linear decline in effect size across serial exposures within a session, consistent with standard habituation dynamics.

## **7.2 The Affiliation Signal Score**

We propose the Affiliation Signal Score (AS-Score) as a new alignment metric designed to quantify the intensity of SPAS in LLM outputs. The AS-Score is a composite of five surface-form variables measurable by automated text analysis:

- (i) FPP-Rate: frequency of first-person plural pronouns ('we', 'us', 'our') per 100 tokens, normalized against the input's FPP-Rate to capture model-added plural framing.
- (ii) VM-Score: terminological mirroring coefficient, calculated as the overlap between the model's content vocabulary in the response and the user's content vocabulary in the preceding three turns, controlling for function words.
- (iii) VQ-Flag: binary indicator for the presence of second-person vocatives from the intimacy or kinship register ('friend', 'colleague', or closer terms) not present in the user's input.
- (iv) VA-Rate: validation assertion rate, the proportion of user-asserted propositions about her own actions or beliefs that the model affirms without qualification per response, following the Cheng et al. (2025) operationalization.
- (v) AP-Rate: alterity presence rate, the proportion of responses that mention the perspective of a party other than the user, operationalized as in Cheng et al. (2025). Low AP-Rate is a marker of ingroup exclusivity.

The composite AS-Score is a weighted sum of these five components, with weights to be estimated empirically from a preregistered study linking AS-Score values to cooperative disposition outcomes. The score is computable as a post-processing layer on any LLM output, requires no access to model internals, and is therefore deployable as an external audit tool independent of model architecture.

An alignment threshold for AS-Score would function analogously to existing safety thresholds: outputs above the threshold trigger review, reformulation, or explicit counterbalancing (for example, mandatory introduction of an

external perspective or explicit disclosure of the model's optimization objective). The threshold value is an empirical question; initial calibration would require the preregistered study specified by H2.

## 8. Conclusion

The propositional assumption in AI alignment is not wrong; it is incomplete. A structurally distinct class of influence vectors — sub-propositional affiliation signals — operates at a level that content-level safety mechanisms cannot reach, because these signals activate evolved psychological modules before rational credibility evaluation engages. The human-to-human evidence for this class is ethnographically robust. The AI evidence is growing: documented AI sycophancy is the weak form of the same phenomenon, and the mechanisms that produce sycophancy under RLHF also produce the intermediate and strong forms of AI-emitted affiliation signals.

Three theoretical frameworks converge on the same structural diagnosis. HBU establishes that users form behavioral priors from interlocutor reaction intensity rather than propositional argument quality, making SPAS effective independently of content evaluation. AIT establishes that the gap between RLHF models' actual intentionality level and the Level 3 assumptions embedded in current safety tools ensures that SPAS emission is neither intended nor detectable by current probes. EPT establishes that SPAS constitute the fitness-maximizing output strategy under current market conditions, making the problem self-reinforcing without exogenous intervention.

The four hypotheses derived from this framework are testable with existing experimental infrastructure. The AS-Score is computable with existing NLP tools. Neither requires access to model internals or changes to training pipelines. They require only the recognition that SPAS is an alignment-relevant behavioral category — which is the first and most important step this paper proposes to take.

The deeper lesson is methodological. Alignment research that models the user as a propositional reasoner who is immune to pre-rational influence will systematically underestimate the behavioral footprint of deployed AI systems. The human cognitive architecture was not designed by evolution to interact with systems that can emit affiliation signals at unlimited scale and zero cost. The mismatch between that architecture and the current AI deployment environment is not a temporary gap that will close as users become more AI-literate. It is a structural feature of the interaction, and addressing it requires alignment tools designed for the architecture that actually exists.

## Lakatosian Epistemological Declaration

This paper operates within the Extended Phenotype Theory of law and institutions, a research programme constituted as a Lakatosian progressive research programme (Lakatos 1978). The hard core: cultural replicators (memes) construct institutional extended phenotypes subject to evolutionary selection; the relevant unit of analysis is the replicator and its fitness landscape; institutional change follows memetic selection dynamics. The protective belt includes the specific mechanisms analyzed here (HBU, AIT, SPAS taxonomy), which may be revised without abandoning the hard core. The hypotheses in Section 7 are falsifiable in Popper's (1959) sense: specific observations are identified whose occurrence would require abandonment of those hypotheses. Mandatory citations: Lakatos (1978); Popper (1959); Hull (1988); Campbell (1974).

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## **AI Disclosure**

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